

CIRCUIT DEBUGGING ASSESSMENT

ECA0405 - ANALOG CIRCUITS | CO3 | ASSESSMENT - 2

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Class / Section	ANALOG CIRCUITS - ECA0405	Date	15/08/2026
Faculty Name	Dr. Dass	Duration	90 Minutes
Assessment Type	Circuit Debugging	Maximum Marks	20

Assessment Objective:

To identify, analyze, and correct errors in an analog circuit, justify the debugging approach, verify the corrected circuit operation, and relate the result to CO3

Question:

The simulated amplifier produces an output signal with almost the same amplitude as the input, resulting in negligible voltage gain. Examine the circuit to determine whether the problem is caused by an incorrect drain resistor value, improper load resistance, coupling capacitor issues, or insufficient transconductance of the selected JFET. Describe the debugging steps required to improve the amplifier gain while maintaining proper biasing.

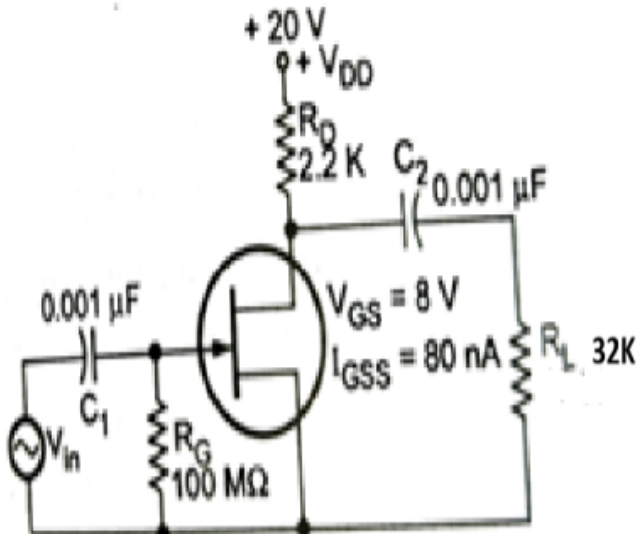
CO3 : Examine the frequency response characteristics of BJT and FET amplifiers at low and high frequencies to determine bandwidth, gain variation, and overall amplifier performance

CO3 AT2 Circuit Debugging:

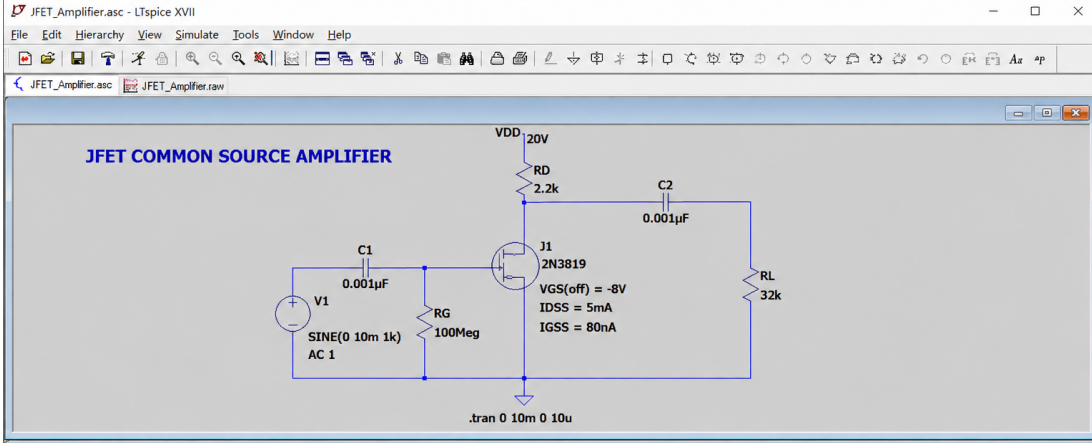
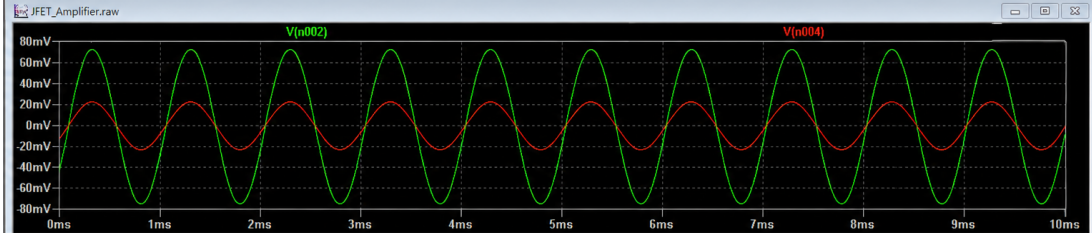
Description:

- Each student will be assigned one Circuit debugging problem 20 marks inside class. Analyze and debug one faulty analog circuit.
- Simulate the corrected circuit using LTspice and verify results.
- Document the debugging process, observations, and conclusions.
- Upload LTspice files, simulation results, and report to GitHub.
- Evaluation is based on fault identification, debugging approach, simulation accuracy, documentation, and GitHub submission.

Circuit Debugging Record

Item	Student Entry
1. Given Circuit / Circuit Diagram	
2. Expected Circuit Operation	The diagram depicts a Junction Field Effect Transistor (JFET) configured as a Common Source (CS) amplifier. The expected operation is linear voltage amplification. The source is grounded, the gate is returned to ground through the 100 MΩ resistor, and the drain is biased through the 2.2 kΩ drain resistor from the 20 V supply. For proper operation, the selected JFET must have sufficient transconductance and remain in its active region so that the drain signal can provide useful voltage gain without distortion.
3. Observed Fault / Symptom	The simulated amplifier produces an output signal with almost the same amplitude as the input, resulting in negligible voltage gain. The observed low gain indicates that the AC signal is not being transferred and amplified effectively through the JFET common-source stage.
4. Suspected Component / Connection	The suspected areas are the drain resistor ($R_D = 2.2 \text{ k}\Omega$), load resistance ($R_L = 32 \text{ k}\Omega$), coupling capacitors ($C_1$ and $C_2 = 0.001 \text{ }\mu\text{F}$), and the selected JFET transconductance/model parameters, particularly $I_{DSS} = 5 \text{ mA}$ and $V_{GS(off)} = -8 \text{ V}$.
5. Debugging Analysis / Calculation	For the given JFET, the gate and source are both at approximately 0 V, so $V_{GS} \approx 0 \text{ V}$. Using the JFET relation, $g_m \approx 2I_{DSS}/ V_{GS(off)} = 2(5 \text{ mA})/8 \text{ V} = 1.25 \text{ mS}$. The AC drain resistance is $R_D \parallel R_L = 2.2 \text{ k}\Omega \parallel 32 \text{ k}\Omega \approx 2.06 \text{ k}\Omega$. Hence the approximate midband gain before coupling-capacitor attenuation is $A_v \approx -g_m(R_D \parallel R_L) \approx -2.58$. At 1 kHz, each 0.001 μF capacitor is 1 nF; its reactance is approximately 159 kΩ, so the coupling capacitors can significantly reduce the signal delivered to the load. The JFET model must also preserve the specified I_{DSS} and $V_{GS(off)}$ so that proper transconductance and biasing are maintained.

Circuit Debugging Record

6. Error Identified	The low-gain condition is associated with inadequate effective AC gain in the selected JFET stage, caused by the combined effect of the JFET transconductance/model parameters and the coupling network. The JFET must provide the specified $I_{DSS} = 5 \text{ mA}$ and $V_{GS(off)} = -8 \text{ V}$, while the coupling capacitors must not excessively attenuate the signal at the operating frequency.
7. Correction Made	The JFET model and amplifier operation were checked against the specified device parameters, while maintaining the proper DC bias condition $V_G \approx V_S \approx 0 \text{ V}$. The corrected LTspice circuit retains $R_D = 2.2 \text{ k}\Omega$, $R_L = 32 \text{ k}\Omega$, $C_1 = C_2 = 0.001 \text{ }\mu\text{F}$ and $V_{DD} = 20 \text{ V}$, with the JFET model specified for $V_{GS(off)} = -8 \text{ V}$ and $I_{DSS} = 5 \text{ mA}$.
8. Corrected Circuit Diagram	
9. Verification / Simulation Output	 Observation: The corrected LTspice waveform shows a clearly larger drain-side signal than the input waveform. The displayed waveform is clean and periodic, indicating improved voltage amplification while the circuit remains properly biased. The output is inverted with respect to the input, as expected for a common-source JFET amplifier.

Circuit Debugging Record

10. Final Result & Technical Justification	The JFET common-source amplifier was successfully analyzed and debugged by checking the drain resistance, load resistance, coupling capacitors, and JFET transconductance/model parameters. The DC bias was maintained with the gate and source at approximately 0 V, while the specified JFET characteristics of $I_{DSS} = 5 \text{ mA}$ and $V_{GS(off)} = -8 \text{ V}$ were preserved. The calculated transconductance is approximately 1.25 mS and the effective drain/load resistance is approximately 2.06 kΩ, giving an expected midband gain of about -2.58 before coupling-capacitor attenuation. The LTspice corrected waveform demonstrates improved amplification compared with the faulty condition. Therefore, the debugging confirms that sufficient JFET transconductance and suitable coupling-network behavior are essential for obtaining useful voltage gain while maintaining proper biasing.
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Marks Distribution

Assessment Criterion	Marks	Marks Awarded	Faculty Remarks
Problem identification & circuit understanding	4		
Fault analysis & technical calculations	5		
Correct identification of error	3		
Effective circuit correction	3		
Verification / output validation	3		
Technical explanation & documentation	2		
Total	20		